

TOOL FAMILIARIZATION

1. Wireshark (Packet Capture)

Introduction

Wireshark is a network protocol analyzer used to capture and inspect network traffic in real time. It helps network administrators and security professionals monitor, troubleshoot, and analyze network communications.

Features

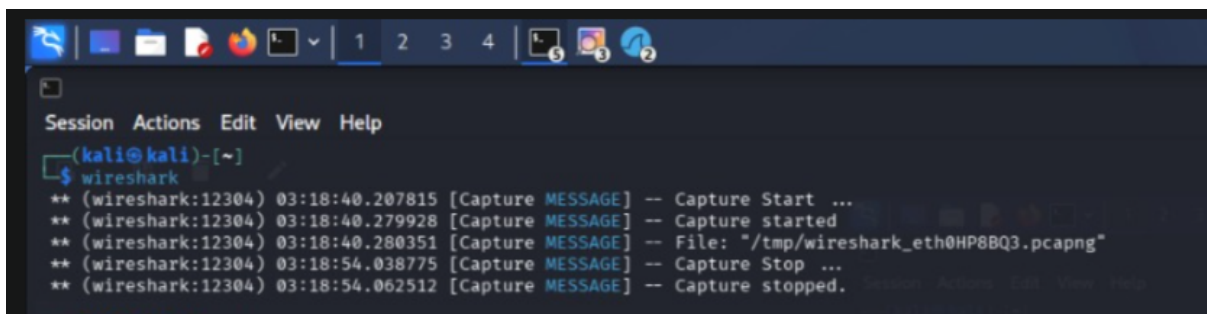
- Captures live network packets
- Analyzes network protocols
- Troubleshoots network issues
- Monitors network traffic
- Supports hundreds of protocols

Practical Activity

1. Opened Wireshark in Kali Linux.
2. Selected the active network interface.
3. Started packet capture.
4. Observed network packets such as TCP, UDP, and DNS traffic.
5. Stopped the capture and analyzed packet details.

Observation

Wireshark successfully captured network packets and displayed detailed information about network communications.



```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ wireshark
** (wireshark:12304) 03:18:40.207815 [Capture MESSAGE] -- Capture Start ...
** (wireshark:12304) 03:18:40.279928 [Capture MESSAGE] -- Capture started
** (wireshark:12304) 03:18:40.280351 [Capture MESSAGE] -- File: "/tmp/wireshark_eth0HP8BQ3.pcapng"
** (wireshark:12304) 03:18:54.038775 [Capture MESSAGE] -- Capture Stop ...
** (wireshark:12304) 03:18:54.062512 [Capture MESSAGE] -- Capture stopped.
```

2. Nmap (Network Scanning)

Introduction

Nmap (Network Mapper) is an open-source tool used for network discovery and security auditing. It helps identify hosts, services, and open ports on a network.

Features

- Host discovery
- Port scanning
- Service detection
- OS detection
- Security auditing

Practical Activity

Command Executed: `nmap localhost`

Observation

Nmap successfully scanned the local machine and displayed information about available network services and open ports.

```
(kali@kali)-[~]
$ nmap localhost
Starting Nmap 7.99 ( https://nmap.org ) at 2026-06-06 03:20 -0400
Nmap scan report for localhost (127.0.0.1)
Host is up (0.0000030s latency).
Other addresses for localhost (not scanned): ::1
All 1000 scanned ports on localhost (127.0.0.1) are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Nmap done: 1 IP address (1 host up) scanned in 0.07 seconds
```

3. Burp Suite (Web Proxy)

Introduction

Burp Suite is a web application security testing platform used for analyzing and testing web applications. It is commonly used in penetration testing.

Features

- Intercepts HTTP/HTTPS requests
- Web application testing
- Vulnerability assessment
- Request modification

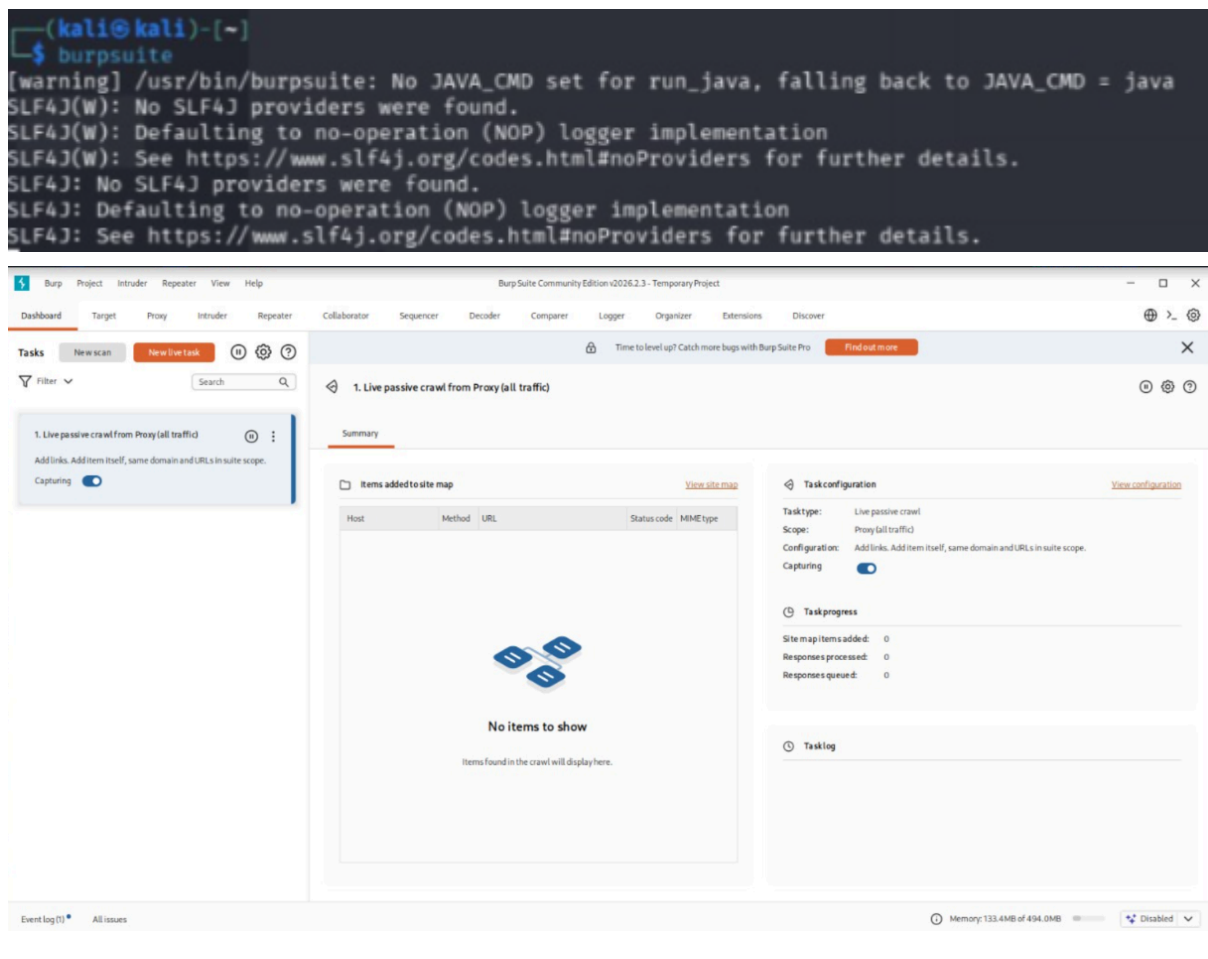
- Traffic analysis

Practical Activity

1. Opened Burp Suite in Kali Linux.
2. Created a temporary project.
3. Accessed the Proxy tab.
4. Explored request interception features.

Observation

Burp Suite successfully launched and provided tools for monitoring and analyzing web application traffic.



4. Netcat (Network Debugging)

Introduction

Netcat is a command-line networking utility used for reading and writing data across network connections. It is often called the “Swiss Army Knife” of networking.

Features

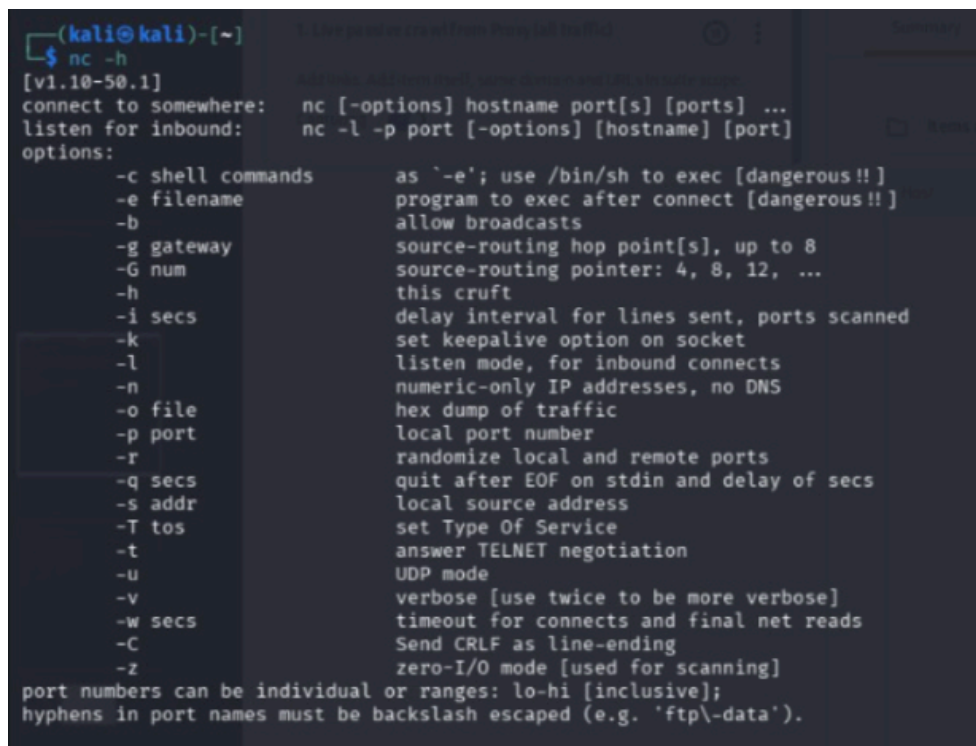
- Network troubleshooting
- Port listening
- Data transfer
- Connectivity testing
- Debugging network services

Practical Activity

Command Executed : nc -h

Observation

Netcat displayed available commands and options used for network debugging and connectivity testing.



```
(kali@kali)-[~]
$ nc -h
[v1.10-50.1]
connect to somewhere: nc [-options] hostname port[s] [ports] ...
listen for inbound: nc -l -p port [-options] [hostname] [port]
options:
  -c shell commands          as '-e'; use /bin/sh to exec [dangerous!!]
  -e filename                program to exec after connect [dangerous!!]
  -b                          allow broadcasts
  -g gateway                 source-routing hop point[s], up to 8
  -G num                     source-routing pointer: 4, 8, 12, ...
  -h                          this cruft
  -i secs                    delay interval for lines sent, ports scanned
  -k                          set keepalive option on socket
  -l                          listen mode, for inbound connects
  -n                          numeric-only IP addresses, no DNS
  -o file                    hex dump of traffic
  -p port                    local port number
  -r                          randomize local and remote ports
  -q secs                    quit after EOF on stdin and delay of secs
  -s addr                    local source address
  -T tos                     set Type Of Service
  -t                          answer TELNET negotiation
  -u                          UDP mode
  -v                          verbose [use twice to be more verbose]
  -w secs                    timeout for connects and final net reads
  -C                          Send CRLF as line-ending
  -Z                          zero-I/O mode [used for scanning]
port numbers can be individual or ranges: lo-hi [inclusive];
hyphens in port names must be backslash escaped (e.g. 'ftp\-data').
```

Conclusion

Wireshark, Nmap, Burp Suite, and Netcat were successfully studied and tested in Kali Linux. These tools play an important role in cybersecurity, network analysis, penetration testing, and troubleshooting. Through this activity, practical exposure was gained to packet analysis, network scanning, web proxy testing, and network debugging techniques.

